

Selecting the “best” model

Readings for today

- Chapter 6: Linear model selection and regularization. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). An introduction to statistical learning: with applications in R (Vol. 6). New York: Springer

Topics

1. How to compare models
2. Full subset selection
3. Stepwise selection

How to compare models

Curse of dimensionality

Dimensions of a model

n: number of observations (i.e., rows)

p: Number of features/independent variables (i.e., columns)

Dimensionality of a model $n \times p$

As $n \rightarrow p$, dimensionality increases & model variance increases

$$\begin{pmatrix} x_{1,1} \\ \dots \\ x_{n,1} \end{pmatrix} \rightarrow \begin{pmatrix} x_{1,1} & x_{1,2} & \dots & x_{1,15} \\ \dots & & & \\ x_{n,1} & x_{n,2} & \dots & x_{n,15} \end{pmatrix} \rightarrow \uparrow \text{model flexibility}$$

Problems from high dimensionality

1. Prediction accuracy: $\uparrow \text{variance} = \downarrow \text{test accuracy}$

- $p > n$ = no unique solution

2. Model interpretability: $\uparrow \text{variance} = \downarrow$ ability to interpret specific $X \rightarrow Y$ relationships

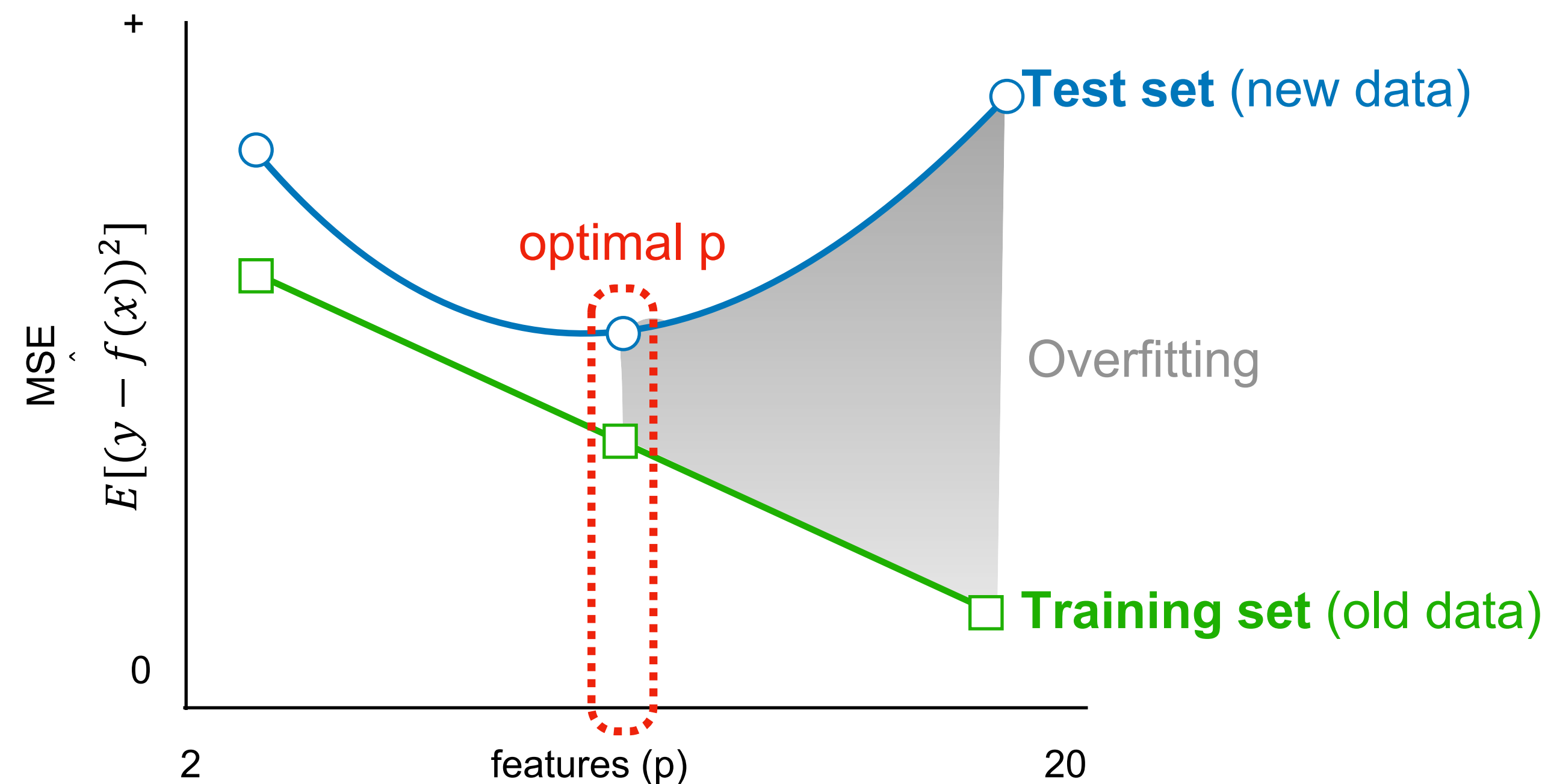
- Feature selection: determine which predictor variables matter

Comparing models

Problem: How do you find the most parsimonious model?

Solutions

1. Validation: Use test error to guide choice of predictors



Comparing models

Problem: How do you find the most parsimonious model?

Solutions

1. Validation: Use test error to guide choice of predictors

2. Bias adjustment: Adjust the *training* error estimate to account for differences in p (i.e., dimensionality)

Bias adjusted criteria

1. Mallow's Cp

$$C_p = \frac{1}{n} (RSS + 2p\sigma_y^2)$$

$$RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

2. Akaike Information Criterion (AIC)

$$AIC = \frac{1}{n\sigma_y^2} (RSS + 2p\sigma_y^2)$$

$$= 2p - 2\log(L)$$


$$L = P(Y|X, \beta, \sigma)$$

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3. Bayesian Information Criterion (BIC)

$$BIC = \frac{1}{n} (RSS + \log(n)p\sigma_y^2)$$

$$= p \log(n) - 2 \log(L)$$

4. Adjusted r^2

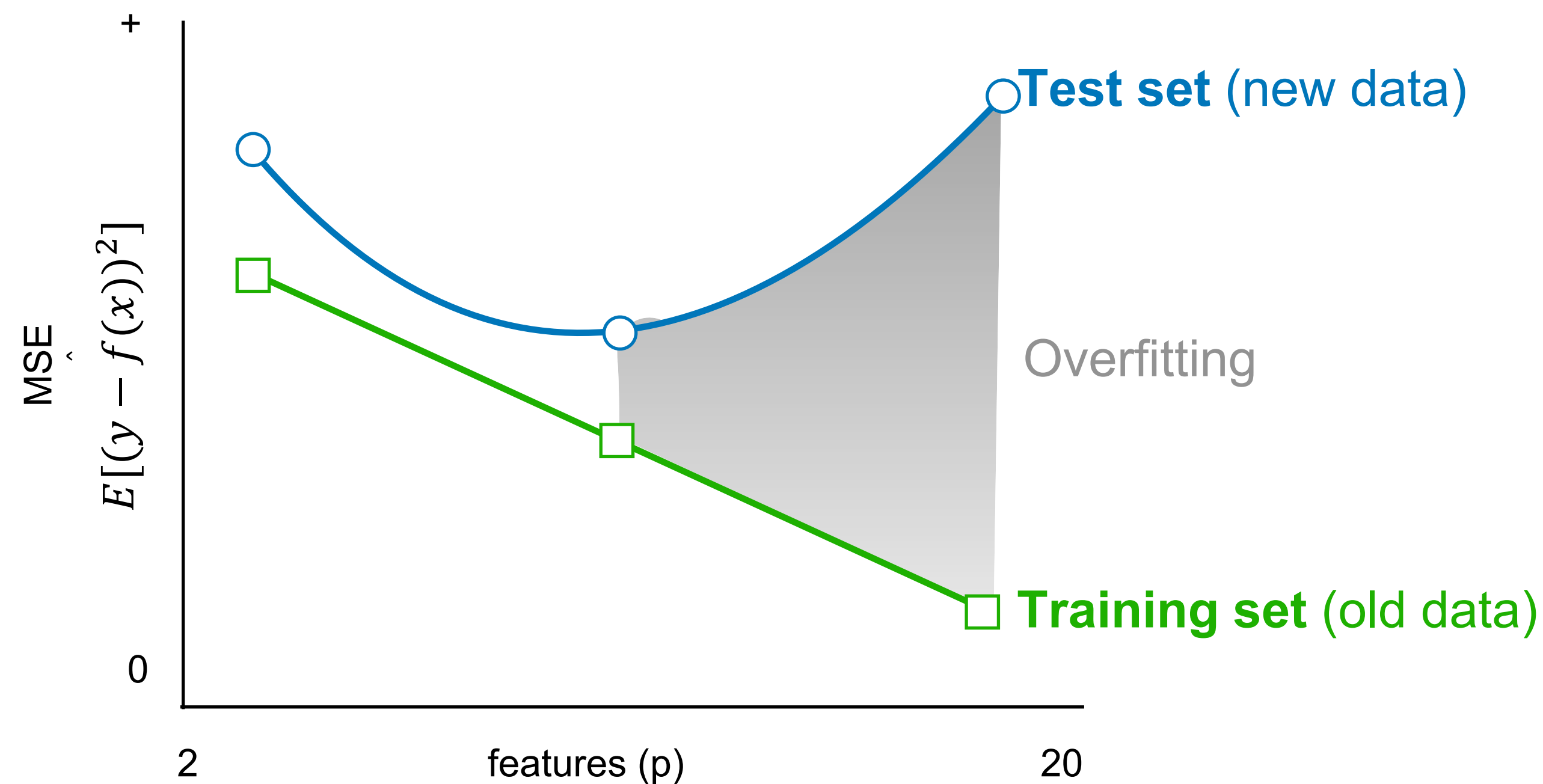
$$adj\ r^2 = 1 - \frac{\frac{1}{n-p-1} RSS}{\frac{1}{n-1} TSS}$$

Full subset selection

What factors are relevant?

Problem: How do you find the most parsimonious model?

$$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \epsilon$$



What is the best combination of variables that provides the best explanation for your data after accounting for model complexity?

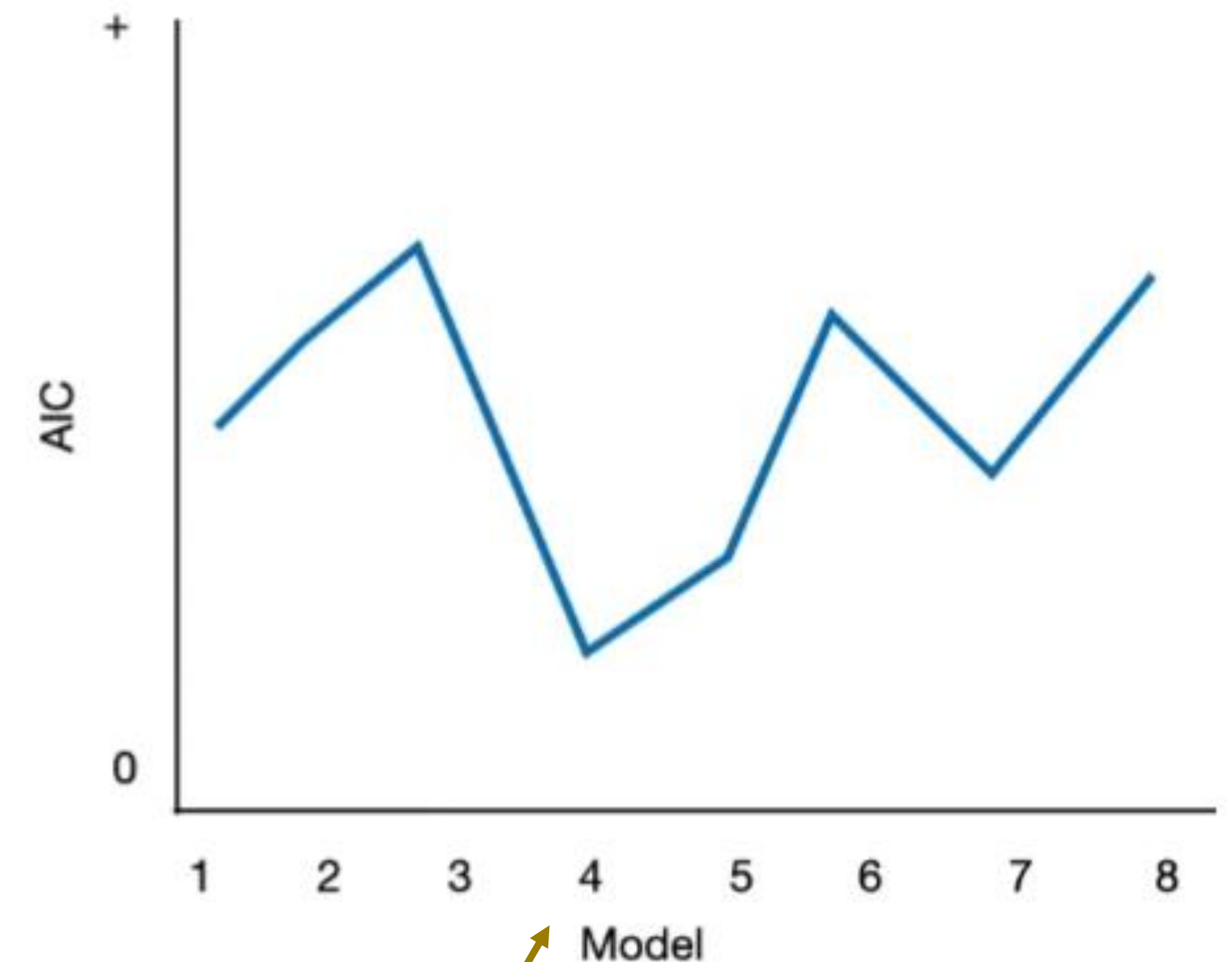
Full subset selection

Goal: Try all possible permutations of your model

eg: $Y = \beta_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \hat{\beta}_3 X_3$

Model variants: 2^p

$Y = \beta_0$	$Y = \beta_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2$
$Y = \beta_0 + \hat{\beta}_1 X_1$	$Y = \beta_0 + \hat{\beta}_1 X_1 + \hat{\beta}_3 X_3$
$Y = \beta_0 + \hat{\beta}_2 X_2$	$Y = \beta_0 + \hat{\beta}_2 X_2 + \hat{\beta}_3 X_3$
$Y = \beta_0 + \hat{\beta}_3 X_3$	$Y = \beta_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \hat{\beta}_3 X_3$



Stepwise selection

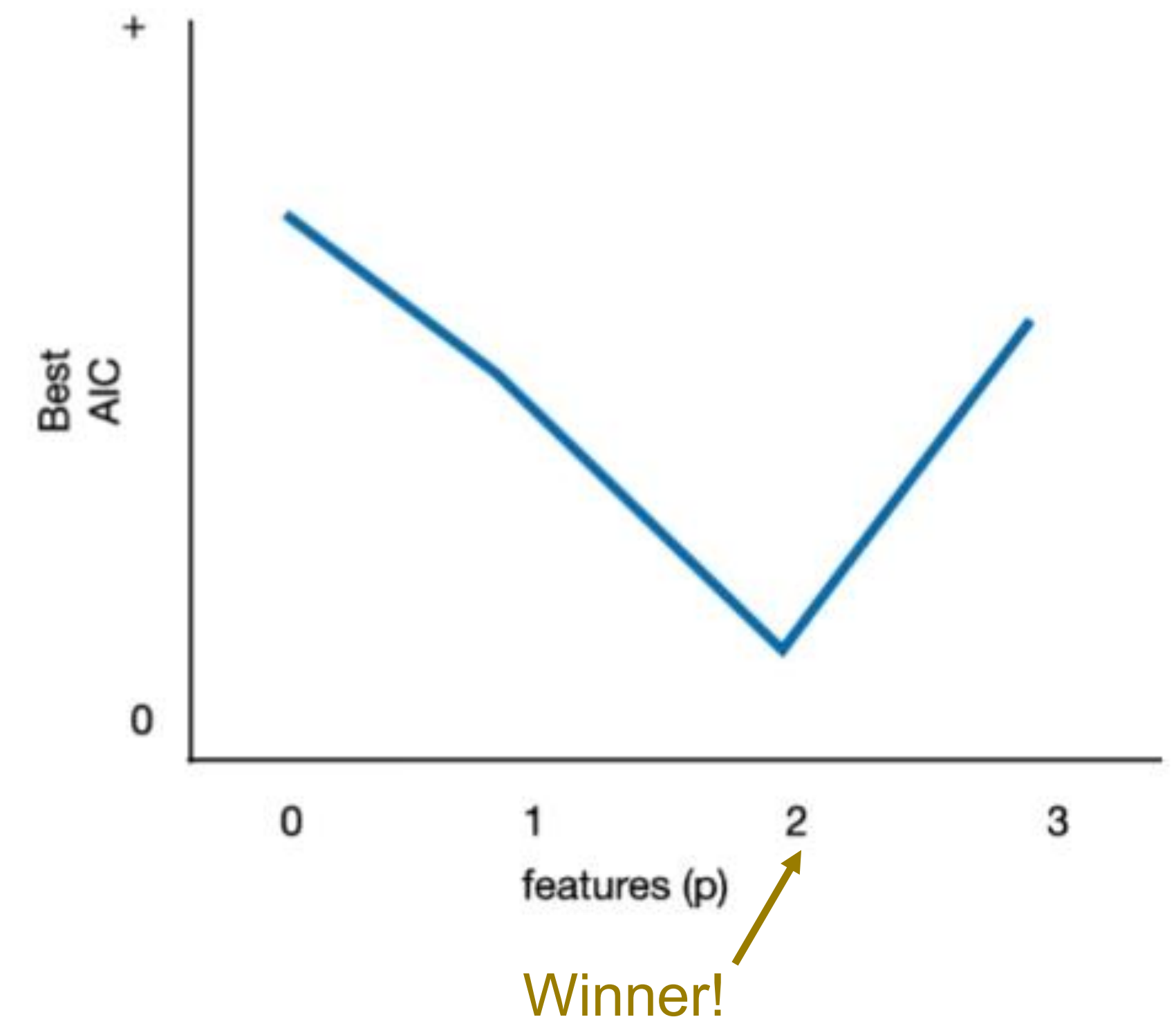
Forward stepwise selection

Goal: Build up from simplest model. Evaluate in stages of same p .

eg: $Y = \beta_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \hat{\beta}_3 X_3$

Model variants: $1 + \frac{p(p+1)}{2}$

- | | |
|--------------------------------------|--|
| 0. $Y = \beta_0$ | 2. $Y = \beta_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2$ |
| 1. $Y = \beta_0 + \hat{\beta}_1 X_1$ | $Y = \beta_0 + \hat{\beta}_1 X_1 + \hat{\beta}_3 X_3$ |
| $Y = \beta_0 + \hat{\beta}_2 X_2$ | 3. $Y = \beta_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \hat{\beta}_3 X_3$ |
| $Y = \beta_0 + \hat{\beta}_3 X_3$ | |



Backwards stepwise selection

Goal: Build down from most complex model. Evaluate in stages of same p .

eg: $Y = \beta_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \hat{\beta}_3 X_3$

Model variants: $1 + \frac{p(p+1)}{2}$

0. $Y = \beta_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \hat{\beta}_3 X_3$

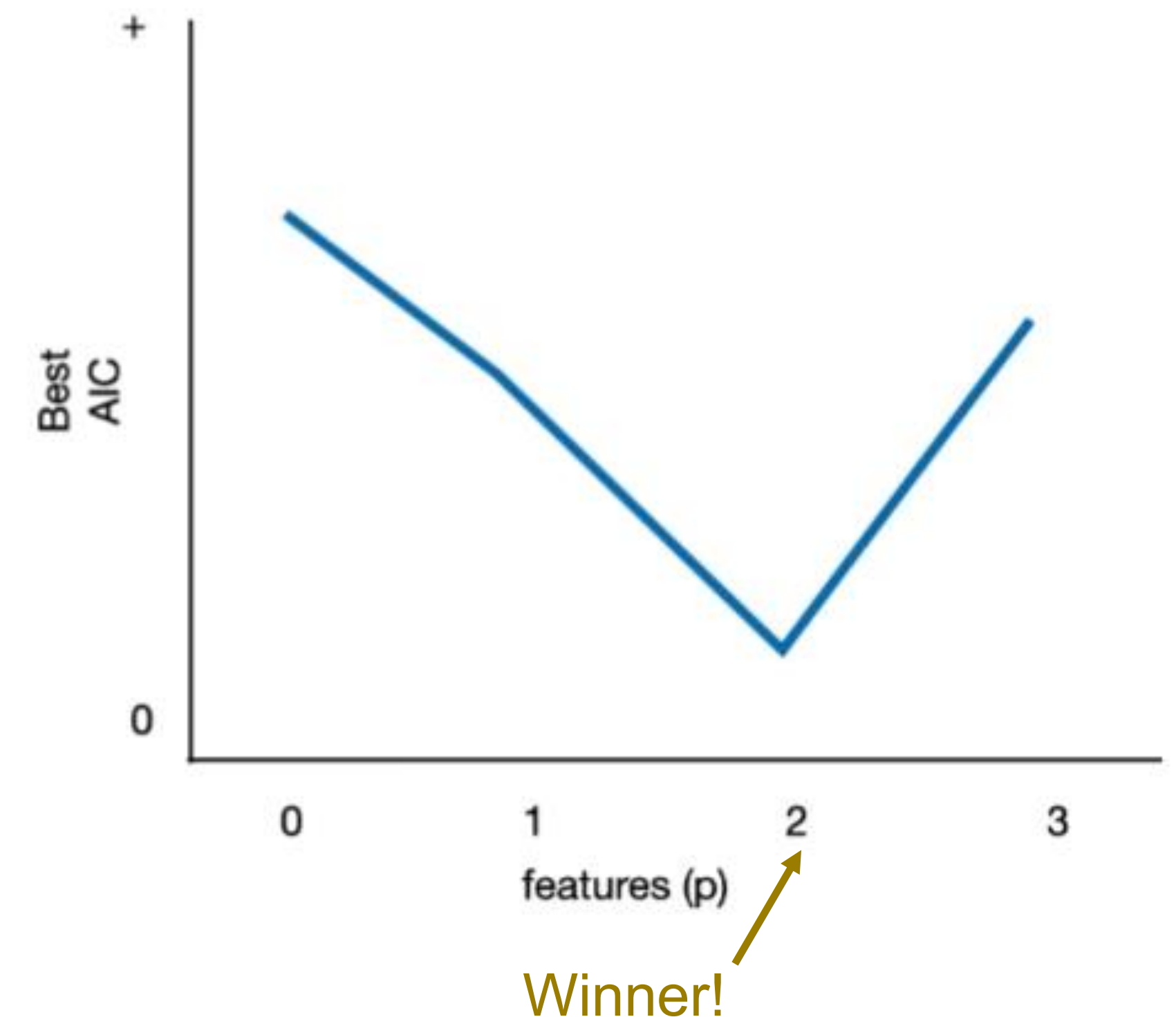
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2. $Y = \beta_0 + \hat{\beta}_1 X_1$
 $Y = \beta_0 + \hat{\beta}_3 X_3$

3. $Y = \beta_0$

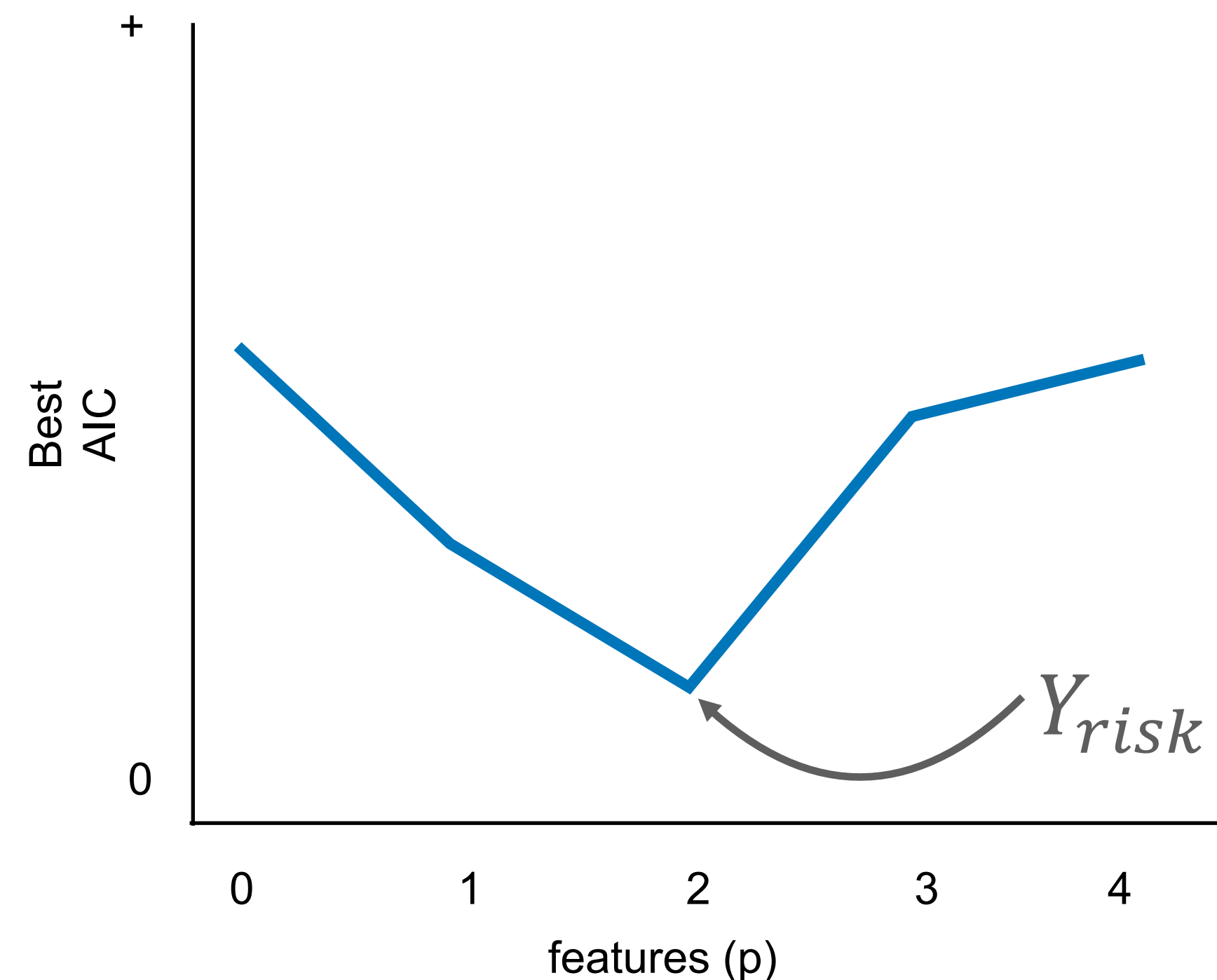


Inference via model selection

Q: Is risk aversion associated with childhood trauma, parental income, psychiatric risk, and/or social network size?

$$Y_{risk} = \hat{\beta}_0 + \hat{\beta}_1 X_{CT} + \hat{\beta}_2 X_{income} + \hat{\beta}_3 X_{psych} + \hat{\beta}_4 X_{social}$$

Conclusion: Only childhood trauma & psychiatric risk are relevant for understanding variation in risk aversion.



$$Y_{risk} = \hat{\beta}_0 + \hat{\beta}_1 X_{CT} + \hat{\beta}_3 X_{psych}$$

Speed vs. completeness

Full subset: 2^p

Stepwise: $1 + \frac{p(p+1)}{2}$

Stepwise methods improve speed dramatically with more complex models.

	p			
	5	10	15	20
Full Subset	32	1,024	32,768	1,048,576
Stepwise	16	56	121	211

} *models compared*

Take home message

- Model selection approaches allow for you to find the most parsimonious at explaining variance of your response variable, Y , while accounting for model complexity.